Project Design Phase
Proposed Solution Template

Date	17 July 2026
Team ID	
Project Name	
Maximum Marks	2 Marks

Project Proposal (Proposed Solution) Report

The proposed project aims to develop an intelligent flood prediction system using Machine Learning to improve disaster preparedness and minimize the impact of floods. The solution analyzes historical rainfall and meteorological data to predict flood risk with high accuracy. By integrating the trained machine learning model into a Flask-based web application, the system enables real-time flood risk prediction, helping individuals and disaster management authorities make informed decisions before flood events occur.

Project Overview

Section	Description
Objective	The primary objective of this project is to develop an AI-powered flood prediction system using Machine Learning that accurately predicts flood risk based on rainfall and weather parameters, enabling early warning and disaster preparedness.
Scope	The project includes data collection, preprocessing, exploratory data analysis, model development, model evaluation, deployment using Flask, and prediction of flood risk through a user-friendly web application. It focuses on improving disaster management and reducing the impact of floods.

Problem Statement

Section	Description
Description	Existing flood warning systems often rely on historical observations and manual analysis, resulting in delayed or inaccurate predictions. The absence of intelligent, localized flood prediction systems makes it difficult for communities and authorities to respond effectively to flood disasters.
Impact	Solving this problem will enable early flood prediction, reduce damage to life and property, improve disaster preparedness, support government agencies in emergency planning, and enhance public safety through timely decision-making.

Proposed Solution

Section	Description
Approach	The proposed solution employs Machine Learning techniques to analyze rainfall and meteorological parameters. Multiple algorithms are evaluated, and the XGBoost model is selected based on its superior performance. The trained model is integrated into a Flask web application to provide real-time flood risk predictions.
Key Features	• Machine Learning-based flood prediction using XGBoost.• Data preprocessing and feature engineering for improved accuracy.• Interactive Flask web application for prediction.• Real-time flood risk assessment based on user input.• High model accuracy with rapid prediction response.• Supports disaster preparedness and informed decision-making.• Scalable architecture for future integration with live weather APIs and GIS mapping.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel Core i5/i7 Processor (or equivalent), 4+ CPU Cores
	Memory	Minimum **8 GB RAM**
	Storage	**512 GB SSD** (Minimum)
Software	Framework	**Flask (Python)**
	Libraries	**XGBoost, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn,**

Resource Type	Description	Specification / Allocation
		Joblib
	Development Environment	**Jupyter Notebook, Visual Studio Code**
Data	Data Source, Size, Format	**Kaggle Flood Prediction Dataset**, CSV format, containing rainfall, meteorological, and flood occurrence attributes